Grade 9 – Mathematics

Third Term Model Paper 5

Chalaka Perera | BSc (Hons) Computer Engineering, University of Peradeniya

Name / Index No.: ..... **Time: 3 hours**

- This paper consists of two parts: Part I and Part II.
- Part I: Answer **all** questions on this paper itself. Part A has 15 questions (2 marks each). Part B has 2 questions (10 marks each).
- Part II: Answer **five** questions only, on separate sheets. Each question carries 10 marks.
- Show all relevant steps of working. Write units where necessary. Use π = 22/7 where needed.

For examiner's use only	
PART I – A	/ 30
PART I – B	/ 20
PART II	/ 50
Total	**/ 100**

PART I

Part A – Answer all questions in the space provided. (2 marks each)

1. Find the 10th term of the number pattern 50, 46, 42, 38, … (2)

Answer: .....

2. Find the value of $11010_2 - 1011_2$ and give the answer in base two. (2)

Answer: .....

3. Write 0.375 as a fraction in its simplest form. (2)

Answer: .....

4. A student scored 42 marks out of 60. Express this as a percentage. (2)

Answer: .....

5. Expand: $(a + 3b)(a – 3b)$ (2)

Answer: .....

6. Simplify: $(a^2b)^3 \div a^4b$ (2)

Answer: .....

7. Round off 47 850 to the nearest thousand. (2)

Answer: .....

8. Solve: $4 – 3x = 2x – 11$ (2)

Answer: .....

9. Describe the locus of points on a plane which are 3 cm from a straight line AB. (2)

Answer: .....

10. In triangle ABC, $\hat{C} = 90°$ and $\hat{A} = 2\hat{B}$. Find $\hat{B}$. (2)

Answer: .....

11. The circumference of a circle is 132 cm. Find its radius. ($\pi = 22/7$) (2)

Answer: .....

12. Can 9 cm, 12 cm and 15 cm be the sides of a right-angled triangle? Give a reason. (2)

Answer: .....

13. Write all the integers x that satisfy $2 < 2x \leq 10$. (2)

Answer: .....

14. n(U) = 30 and n(A) = 18. Find n(A′). (2)

Answer: .....

15. Make r the subject of the formula $C = 2\pi r$. (2)

Answer: .....

Part B – Answer both questions on this paper. (10 marks each)

1.

A point P on level ground is 30 m from the foot F of a vertical tower FT. The angle of elevation of the top T of the tower from P is 40°.

(i) Draw a rough sketch showing this information. (2 marks)

(ii) Using the scale 1 cm = 5 m, draw a scale diagram. (2 marks)

(iii) Using your scale diagram, find the height of the tower. (2 marks)

(iv) Using your scale diagram, find the distance PT. (2 marks)

(v) Write the angle of depression of P from T, giving a reason. (2 marks)

2.

U = {1, 2, 3, …, 12}, A = {multiples of 2} and B = {multiples of 3} are subsets of U.

(i) Write the elements of A and B. (2 marks)

(ii) Represent U, A and B in a Venn diagram. (2 marks)

(iii) Write the elements of $A \cap B$ and $A \cup B$. (2 marks)

(iv) Write the elements of $(A \cup B)'$. (2 marks)

(v) Find $n(A \cap B')$ and describe this set in words. (2 marks)

PART II

Answer five questions only. (10 marks each)

1.

(a) Simplify: $(2\frac{1}{2} - 1\frac{1}{3}) \times 1\frac{1}{5}$ (4 marks)

(b) A water tank is $\frac{3}{8}$ full. When 150 litres of water is added, it becomes $\frac{3}{4}$ full.

(i) What fraction of the tank does 150 litres fill? (2 marks)

(ii) Find the capacity of the tank. (2 marks)

(iii) How much water was in the tank at first? (2 marks)

2.

(a) Simplify: $(2x^3)^2 \times 3x^{-4}$ (3 marks)

(b) Solve: $5^{2x-1} = 125$ (3 marks)

(c) Find the value of $(6.4 \times 10^6) \div (1.6 \times 10^2)$ and give the answer in scientific notation. (2 marks)

(d) Round off 3.14159 (i) to three significant figures, (ii) to four decimal places. (2 marks)

3.

A taxi charges a fixed fee of Rs 100 plus Rs 80 per kilometre. The fare y (in rupees) for a journey of x km is given by $y = 80x + 100$.

x (km)	0	1	2	3	4	5
y (Rs)	100	…	260	…	…	500

(i) Complete the table of values. (2 marks)

(ii) Draw the graph on graph paper (x-axis: 1 cm = 1 km, y-axis: 1 cm = Rs 50). (3 marks)

(iii) Using the graph, find the fare for a journey of 3.5 km. (2 marks)

(iv) What does the gradient of the graph represent? (1 marks)

(v) A passenger has Rs 700. Find the greatest whole number of kilometres he can travel. (2 marks)

4.

(a) Solve: $(x + 3)/4 + (x - 1)/2 = 4$ (3 marks)

(b) An adult ticket for a show costs Rs 150 and a child ticket costs Rs 80. A group bought 20 tickets for Rs 2300. Form a pair of simultaneous equations and find the number of adult tickets and child tickets. (5 marks)

(c) When a whole number is multiplied by 3 and 5 is subtracted, the result is at most 25. Form an inequality and find the largest such number. (2 marks)

5.

Use only a straight edge with a cm/mm scale and a pair of compasses for the following constructions. Show all construction lines.

(i) Draw a straight line segment XY of length 8 cm. (1 marks)

(ii) Construct an angle of 120° at X such that XY is an arm. (3 marks)

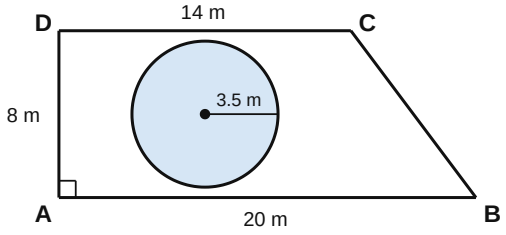
(iii) Mark Z on the arm such that XZ = 5 cm, and complete the triangle XYZ. (1 marks)

(iv) Produce YX and construct a perpendicular from Z to YX produced. Name its foot D. (3 marks)

(v) Measure and write the lengths of ZD and XD. (2 marks)

6.

ABCD is a garden in the shape of a right trapezium, with AB ∥ DC and DÂB = 90°. AB = 20 m, DC = 14 m and AD = 8 m. There is a circular flower bed of radius 3.5 m in the garden. (π = 22/7)



(i) Find the area of the garden. (2 marks)

(ii) Find the area of the flower bed. (2 marks)

(iii) The rest of the garden is to be covered with grass at Rs 80 per m². Find the cost. (3 marks)

(iv) Using Pythagoras' relation, find the length of BC. (2 marks)

(v) Find the perimeter of the garden. (1 marks)

7.

(a) A box contains 20 identical cards numbered 1 to 20. A card is drawn at random. Find the probability that the number on it is

(i) a multiple of 5 (ii) a square number (iii) greater than 15 (iv) not a multiple of 5 (4 marks)

(b) The mean of five numbers is 12. Four of the numbers are 10, 15, 8 and 14.

(i) Find the sum of the five numbers. (ii) Find the fifth number. (iii) If 18 is added as a sixth number, find the new mean. (6 marks)